InveStar

Remit to Invest on Stellar

Technical Architecture & Traction Evidence

$28B	3–5s	~1%	$141K
US–BD corridor/yr	Stellar settlement	Target cost	SCF request

SCF Build Award #43 · **Integration Track (Large)** · $141,325 · 16 Weeks

investarbd.com · hello@investarbd.com · SCF #43

InveStar - Technical Architecture Document

SCF Build Award #43 | Integration Track (Large) | $141,325 | 16 Weeks

May 1 - August 21, 2026 | MoneyGram Ramps | SEP-10/24/12 | Soroban automation T3

Sections 1-10: Architecture | Annex A-D: Traction, Budget, Milestones, Regulatory

Section 1 - System Overview

InveStar replaces SWIFT with Stellar USDC rails for the US->Bangladesh corridor, then layers a new financial primitive: every remittance can automatically route a user-defined percentage into Dhaka Stock Exchange equities via InveStar's live B2B DSE OMS — direct brokerage connections, live today. Soroban smart contracts (T3) will automate this routing fully on-chain. No incumbent (Wise, TapTap, Remitly) combines Stellar settlement + DSE investing.

> Why Stellar: MoneyGram Ramps (SCF Integration List building block) runs exclusively on Stellar SEP-24/10. Bangladesh Bank USDC prohibition is solved by the SEP-12/24 compliance stack with MoneyGram FX infrastructure. Stellar's $0.001 tx fee is the only way to hit ~1% all-in cost on a $50-200 transfer. Soroban smart contracts (T3) will automate the remit-to-invest routing on-chain — Stellar-native, no equivalent on Ethereum or Solana.

Five Backend Modules - All Built Around Stellar

Mod.	Service Name	Stellar Integration	Status
A	stellar-ramp-service	SEP-10/24 | Horizon API | AES-256-GCM/HSM key custody | USDC SAC	Live - 993 wallets | 1,000+ ops
B	moneygram-transfer-service	USDC settlement event -> MG Transfer API | bKash/bank BDT delivery	Sandbox tested | MG partner active
C	treasury-conversion-service	USDC SAC queries | MoneyGram FX bridge (BD Bank compliance)	Architecture complete
D	remit-to-invest-connector	Horizon SSE -> InveStar B2B DSE OMS (live) | Soroban AutoInvestVault automation planned T3	DSE OMS live | Soroban T3
E	ai-coach-service	Horizon API read-only | tx history | fee savings display	Excluded from budget

Section 2 - C4 Level 1: System Context

Shows InveStar as a black box with its full relationships to external actors and all external systems.

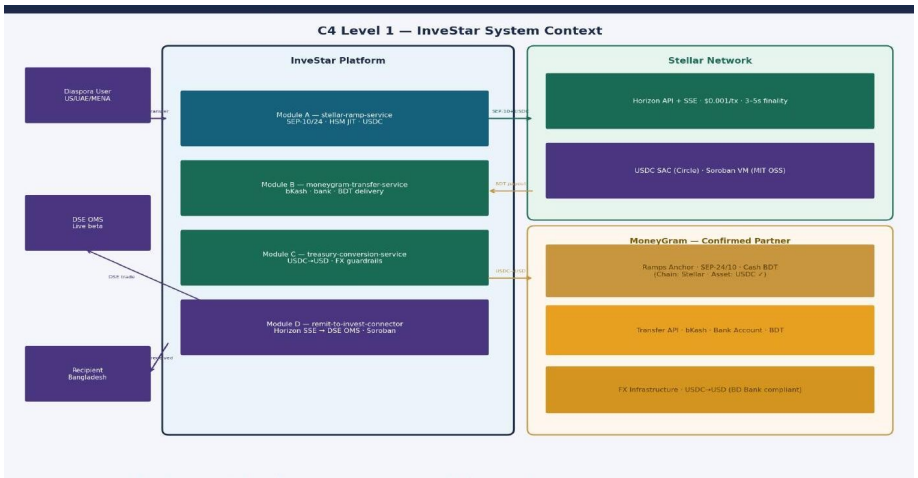


Figure 1: C4 Level 1 - US/UAE/MENA sender -> Stellar USDC -> MoneyGram (confirmed partner, Chain: Stellar, Asset: USDC) -> BDT recipient

Section 3 - C4 Level 2: Container Architecture

Each module maps directly to a funded SCF milestone.

Figure 2: C4 Level 2 - Container Architecture | Mod A=M1.1-M1.2 | Mod B=M2.2 | Mod C=M2.1 | Mod D=M3.2-M3.3 | Azure Key Vault HSM FIPS 140-2 L3

Layer	Containers & Technology
Client Layer	Web App (Next.js TypeScript live) | Mobile App (React Native T3 new) | DSE OMS UI (React live)
API Gateway	NestJS | Azure APIM | SEP-10 JWT | MFA | Rate limiting | Redis
Backend Services	Mod A: stellar-ramp-service | Mod B: moneygram-transfer-service | Mod C: treasury-conversion-service | Mod D: remit-to-invest-connector
Data Layer	PostgreSQL (Supabase RLS) | Redis | Azure Blob (KYC) | Azure Key Vault HSM (FIPS 140-2 L3)

Stellar Rail	Horizon API | USDC SAC (Circle) | SEP-10/24/12 | Soroban VM (AutoInvestVault — planned T3, MIT OSS)
Ext. Partners	MoneyGram Ramps (confirmed) | MoneyGram Transfer API + FX | DSE OMS API (live)

Section 4 - Stellar Integration: Deep Technical Detail

USDC Settlement - How Stellar SCP Works

USDC on Stellar is a native Stellar Asset Contract (SAC) issued by Circle - not a wrapped token. No bridge risk. SCP reaches federated Byzantine agreement in 3-5 seconds at $0.001. Settlement is immediate and irreversible after ledger close.

```
// USDC on Stellar - native SAC, NOT a wrapped token
// Issuer: GA5ZSEJYB37JRC5AVCIA5MOP4RHTM335X2KGX3IHOJAPP5RE34K4KZVN | 993 wallets | 1,000+
ops
const buildUsdcTransfer = async (sourceAccount, destination, amount) => {
  const usdcAsset = new Asset("USDC", CIRCLE_USDC_ISSUER);
  return new TransactionBuilder(sourceAccount, {
    fee: BASE_FEE,                      // 100 stroops = $0.000001
    networkPassphrase: Networks.PUBLIC, // Stellar MAINNET
  }).addOperation(Operation.payment({ destination, asset: usdcAsset, amount }))
    .setTimeout(30).build();
  // After signing -> submitTransaction() -> SCP quorum | Finality: 3-5s | Irreversible
};
```

SEP-10 Authentication with MoneyGram Ramps Anchor

```
// stellar-ramp-service - SEP-10 JWT challenge/response
// Account: GAGZEFSXEER5ZDIAXLEHJ7MFTZRTYWPCKU3SBQI6Z5GJLIYCBMY7XVNM
async authenticateWithAnchor(userPublicKey) {
  // 1. Fetch challenge from MG Ramps anchor (confirmed partner)
  const { transaction } = await fetch(`${MG_RAMPS_ANCHOR_URL}/auth?account=$
{userPublicKey}`).then(r=>r.json());
  // 2. Sign inside key custody (AES-256-GCM staging | Azure HSM production)
  const signedTx = await this.custodyService.signTransaction(tx);
  // 3. Return signed challenge -> anchor issues JWT
  const { token } = await fetch(`${MG_RAMPS_ANCHOR_URL}/auth`, {
    method: "POST", body: JSON.stringify({ transaction: signedTx.toEnvelope().toXDR("base64")
}),
  }).then(r=>r.json());
  return token; // JWT for all subsequent SEP-24/12 calls
}
```

The diagram below shows the complete Track 1 SEP-24 cash pickup sequence — SEP-10 auth through USDC settlement to BDT pickup:

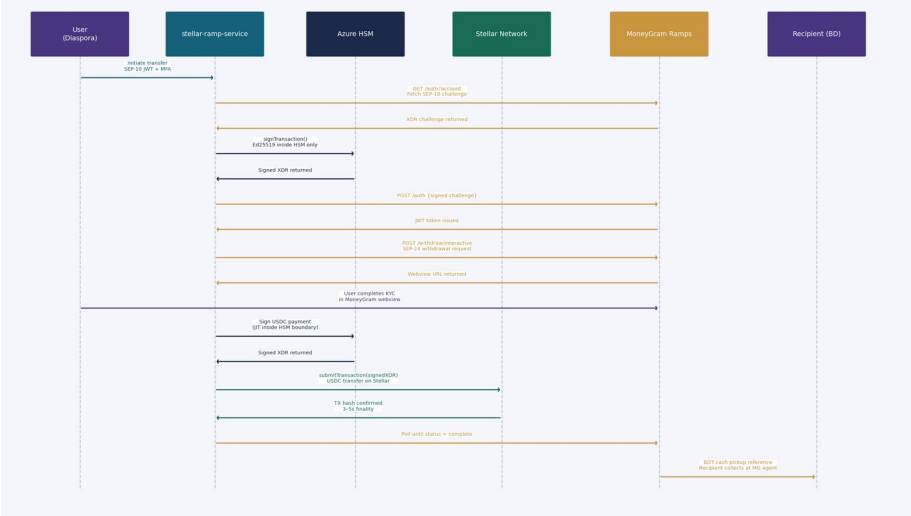


Figure 3a: Track 1 SEP-24 Cash Pickup — full 16-step sequence: SEP-10 JWT auth → HSM JIT signing → USDC on Stellar (3-5s) → MoneyGram anchor → BDT cash pickup (T1 MVP · M1.2)

Horizon SSE Streaming - Real-Time Settlement Listener

```
// remit-to-invest-connector - Horizon SSE (no polling, sub-second delivery)
// Treasury: GCCEMCYSJLY7YRPQ4WJMXDUP3RNCC7F2XOXZAYVULSWPGZWRB5PO67IN
startListening() {
  this.server.payments().forAccount(INVESTAR_TREASURY_ACCOUNT).cursor("now")
    .stream({ onmessage: async (payment) => {
      if (payment.type=="payment" && payment.asset_code=="USDC" &&
          parseFloat(payment.amount) >= MIN_INVEST_THRESHOLD) {
        await this.onSettlement({ txHash: payment.transaction_hash, amount:
payment.amount });
      }
    }});
}
```

When an inbound USDC payment is detected, the Horizon SSE listener triggers the remit-to-invest routing to InveStar's live B2B DSE OMS:

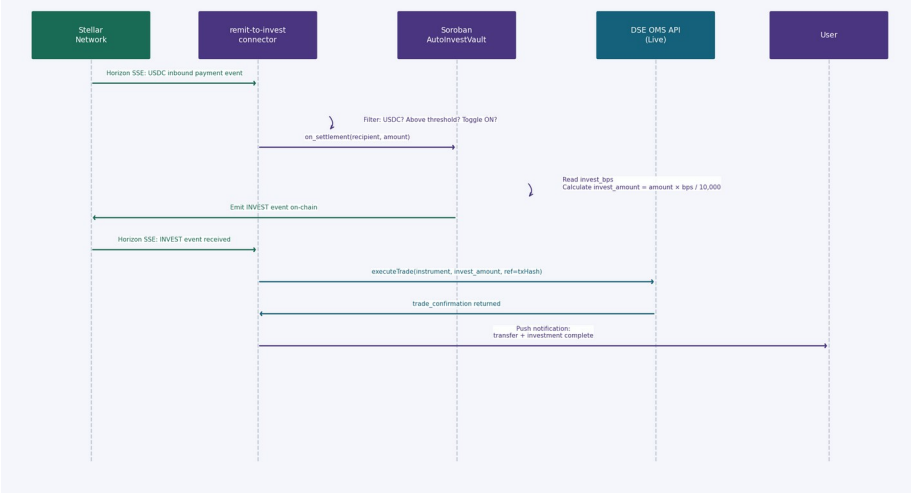


Figure 5a: Horizon SSE → remit-to-invest-connector → InveStar B2B DSE OMS (live, direct brokerage) → DSE trade. Soroban AutoInvestVault (T3 M3.3) will replace the off-chain bridge with on-chain automation.

MoneyGram - Confirmed Partner: Module C Architecture

> InveStar is a confirmed MoneyGram partner (INVESTAR REMIT | InveStar Inc | Chain: Stellar | Asset: USDC | Products: Withdraw, Deposit | Status: Active). MoneyGram serves three roles: (1) SEP-24 Ramps anchor for cash pickup (Track 1), (2) Transfer API for bKash/bank (Track 2), (3) FX conversion for Module C. BD Bank USDC prohibition solved by converting outside Bangladesh. See Annex A for portal screenshot.

```
// treasury-conversion-service - MoneyGram FX (confirmed partner)
// BD Bank prohibits direct USDC receipt - conversion happens OUTSIDE Bangladesh
async convertUsdcToUsd(params) {
  const rate = await this.mgFxOracle.getRate("USDC", "USD");
  if (Math.abs(rate - 1.0) > FX_TOLERANCE) throw new FxGuardrailError(`Rate ${rate} outside
tolerance`);
  const result = await this.moneygramClient.convertUsdcToUsd({
    amount: params.usdcAmount, idempotencyKey: params.stellarTxHash,
  });
  await this.floatAccount.credit(result.usdAmount);
  await this.auditLog.record({ ...params, result, rate });
  return result;
}
```

Track 2 settlement flow — USDC on Stellar triggers the full treasury conversion and MoneyGram bKash/bank payout:

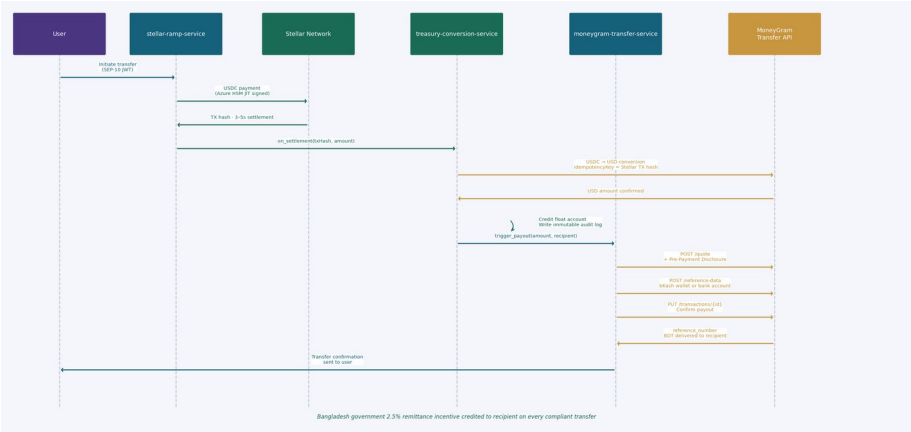


Figure 4a: Track 2 — USDC settles Stellar (3-5s) → treasury-conversion-service (FX guardrail + MoneyGram FX) → moneygram-transfer-service delivers BDT to bKash wallet or bank account (T2 · M2.2)

Stellar vs SWIFT

Dimension	SWIFT / Traditional	InveStar / Stellar
Settlement time	2-4 business days	3-5 seconds
Transaction fee	$35-95	$0.001
Intermediaries	2-4 correspondent banks	Zero
Finality	Reversible T+2	Immediate, irreversible
MoneyGram access	Fiat rails only	SEP-24/10 | Confirmed partner
Programmable investing	Impossible	Live DSE OMS today + Soroban on-chain automation (T3)
All-in cost	5-7% (WB Q1 2025)	~1% target
On-chain wallets	N/A	993 beta-user wallets verified

Section 5 - Data Flows

Track 1 - SEP-24 Cash Pickup (MVP T1)

#	Actor / Service	Action
1	User -> API Gateway	Initiates transfer | SEP-10 JWT verified | MFA gate passed
2	API Gateway -> stellar-ramp-service	Authorised request routed to Module A
3	stellar-ramp-service -> MoneyGram Ramps	SEP-10 challenge fetched, signed inside HSM, JWT issued
4	stellar-ramp-service -> MoneyGram Ramps	POST /transactions/withdraw/interactive | SEP-24 withdrawal | webview URL returned
5	User -> MoneyGram webview	KYC completed inside MoneyGram hosted webview
6	stellar-ramp-service -> Key Custody	USDC payment tx signed JIT | Ed25519 | private key never in app memory
7	stellar-ramp-service -> Stellar Network	submitTransaction(signedXDR) -> SCP consensus | 3-5s finality | TX hash confirmed
8-10	Poll -> Reference -> Recipient	Poll until status=completed | Cash pickup reference | Recipient collects BDT at MoneyGram agent

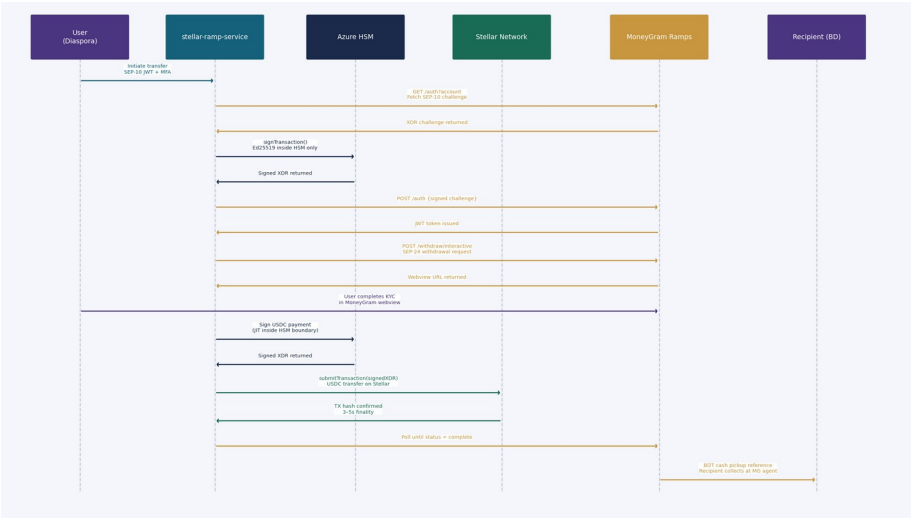


Figure 3: Track 1 SEP-24 Cash Pickup - T1 MVP | Milestone 1.2

Track 2 - MoneyGram bKash / Bank Payout (T2)

#	Service	Action
1	stellar-ramp-service -> Stellar	USDC payment submitted | 3-5s settlement | TX hash confirmed
2	treasury-conversion-service	FX guardrail: USDC/USD rate within tolerance | Daily position limit checked
3	treasury-conversion-service ->	USDC->USD via MoneyGram FX

	MoneyGram	(confirmed partner) | idempotencyKey = Stellar TX hash
4	treasury-conversion-service	USD credited to float account | Immutable audit log written
5-7	moneygram-transfer-service -> MG Transfer API	POST /quote | POST /reference-data | PUT /transactions/:id -> BDT delivered to recipient
8	Recipient	BDT received in bKash wallet or bank account | BD govt 2.5% incentive reconciled

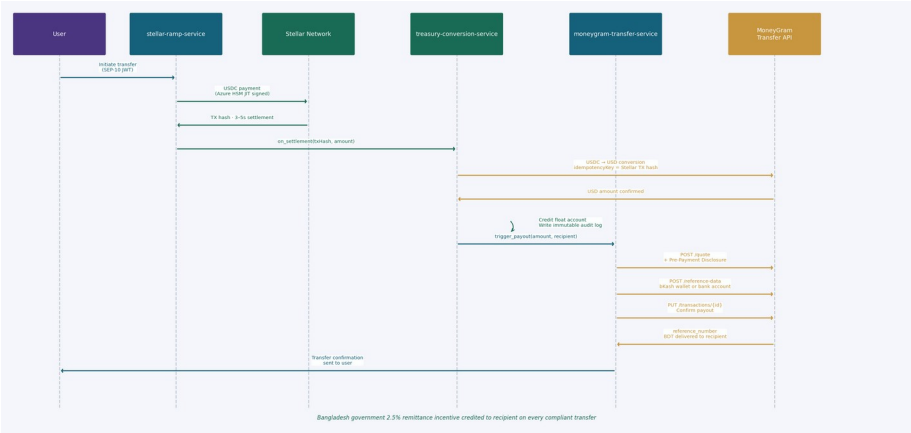


Figure 4: Track 2 - USDC settles Stellar -> treasury converts -> MoneyGram delivers BDT to bKash/bank | T2 | Milestone 2.2

Remit-to-Invest - Stellar Settlement -> DSE Trade (T3)

#	Service	Action
1	Stellar -> remit-to-invest-connector	Horizon SSE: USDC inbound payment event received
2	remit-to-invest-connector	Filter: USDC? Above threshold? User invest toggle ON?
3	connector -> InveStar DSE OMS API	executeTrade(instrument, invest_amount, ref=stellarTxHash) via live B2B OMS — direct brokerage connection to DSE
4	DSE OMS API -> connector	Trade confirmed (DSE reference number returned)
5	connector -> User	Portfolio updated | trade confirmation pushed via app | Soroban (T3) will automate steps 2-4 fully on-chain

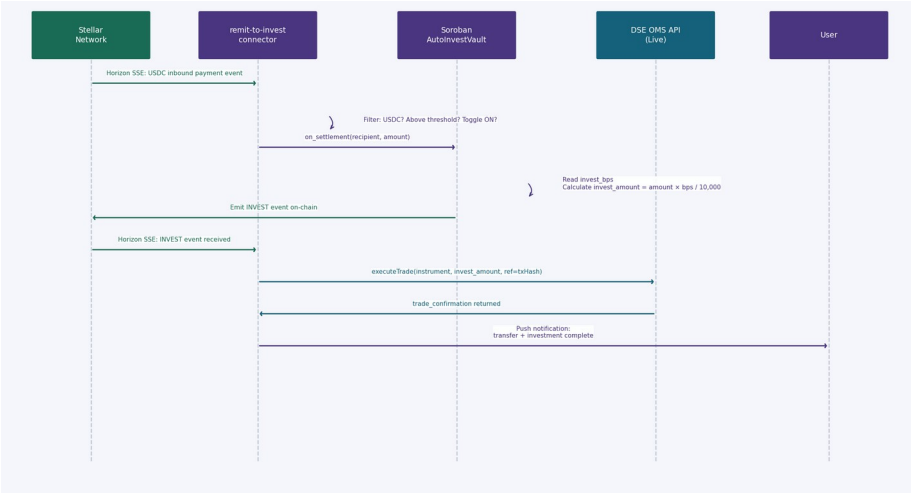


Figure 5: Remit-to-Invest — Horizon SSE detects USDC settlement -> connector routes invest amount to InveStar B2B DSE OMS (live, direct brokerage) -> DSE trade executed. Soroban AutoInvestVault (T3 M3.3) will automate this on-chain.

Section 6 - Security Model

Defence-in-Depth - Five Security Layers

```
▶️ Defence-in-Depth - Five Security Layers
Layer 1 NETWORK: TLS 1.3 all endpoints | Azure APIM rate limiting
     |
Layer 2 AUTH: SEP-10 JWT challenge-response (MG Ramps anchor) | MFA (TOTP/OTP)
     |
Layer 3 KEY CUSTODY:
  AES-256-GCM (Supabase edge fn) - testnet / staging
  Azure Key Vault HSM FIPS 140-2 L3 - production mainnet
  Private key NEVER leaves HSM | Ed25519 JIT signing
     |
Layer 4 APPLICATION:
  Check-Effect-Interaction (double-spend prevention)
  Idempotency: Stellar TX hash as dedup key
  AI Coach hard wall: zero write access, NemoClaw
     |
Layer 5 COMPLIANCE:
  FX guardrail: USDC/USD rate tolerance check
  SEP-12 AML/KYC exchange with MG anchor
  Append-only audit log anchored to Stellar TX hash
```

HSM JIT Signing Architecture

```
// stellar-ramp-service - Azure Key Vault HSM | FIPS 140-2 Level 3
// Private key NEVER exported from HSM - only key reference held by app
async signTransaction(tx) {
  const txHash = tx.hash();
  const { result: signature } = await this.kvClient.sign(
    KEY_VAULT_URI, HSM_KEY_ID, "EdDSA", txHash); // Ed25519 sign INSIDE HSM
  tx.addSignatureDecorated({ hint: this.getPublicKeyHint(), signature: Buffer.from(signature)
});
  return tx;
}
```

Check-Effect-Interaction - Double-Spend Prevention

```
async processTransfer(params) {
  // CHECK: validate before touching state
  const key = sha256(params.userId + params.amount + params.nonce);
  if (await this.cache.exists(key)) throw new DuplicateTransactionError();
  // EFFECT: write PENDING before any external call
  await this.audit.writePending({ ...params, key });
  await this.cache.set(key, "processing", TTL_30S);
  // INTERACTION: external systems only after state is safe
  const txHash = await this.stellarRamp.signAndSubmit(params);
  await this.audit.writeConfirmed({ ...params, txHash });
  return { success: true, txHash };
}
```

Section 7 - Soroban Smart Contracts: Planned T3 Deliverable (MIT OSS)

> Current state: InveStar routes invest amounts via its live B2B DSE OMS with direct brokerage connections to the Dhaka Stock Exchange. Soroban AutoInvestVault (T3 Milestone 3.3) will automate this routing fully on-chain. Open-source commitment: All Soroban contracts published under MIT licence before T3 payment is claimed — verifiable via Stellar testnet deployment tx hash.

▶️ Soroban Planned Architecture: Phase 1 (T3) -> Phase 2 -> Phase 3

```
PHASE 1 | T3 | This SCF Round | MIT OSS
  Stellar USDC Settlement (3-5s) -> on_settlement(recipient, amount)
       +-- PreferenceStore.get(recipient) -> invest_bps + instrument
       +-- env.events().publish(INVEST, invest_amount)
                 | Horizon SSE sub-second delivery
                 v remit-to-invest-connector
                 -> executeTrade() -> DSE OMS API (live)


PHASE 2 | Next SCF Round
  InvestRouter (cross-contract) -> DSE Adapter
  VaultFactory: isolated vault per user | Principal Invariant


PHASE 3 | SCF Round +2
  YieldVault: DRR model (B=D*R local | C=D*(1-R) DeFi)
  Blend/DeFindex | APY_eff = APY_DeFi * (1-R)
```

```rust
// AutoInvestVault.rs - Phase 1 | GitHub: public | MIT licence | committed at M3.3
#[contractimpl]
impl AutoInvestVault {
  pub fn on_settlement(env: Env, recipient: Address, amount: i128) {
    recipient.require_auth();
    let pref = env.storage().instance().get(&DataKey::Preference(recipient.clone()))
      .unwrap_or(InvestPreference { invest_bps:0, is_active:false,
instrument:symbol_short("NONE") });
    if !pref.is_active || pref.invest_bps == 0 { return; }
    assert!(pref.invest_bps <= 10_000, "invest_bps cannot exceed 100%");
    let invest_amount = (amount * pref.invest_bps as i128) / 10_000;
    // Emit on-chain event - Horizon SSE delivers sub-second
    env.events().publish((symbol_short("INVEST"), recipient.clone()),
                         (invest_amount, amount - invest_amount, pref.instrument));
  }
  pub fn set_preference(env: Env, user: Address, invest_bps: u32, instrument: Symbol) {
    user.require_auth();
    assert!(invest_bps <= 10_000);
    env.storage().instance().set(&DataKey::Preference(user),
      &InvestPreference { invest_bps, instrument, is_active: invest_bps > 0 });
    env.storage().instance().extend_ttl(17_280, 17_280); // ~1 day TTL
  }
}
```

Section 8 - Soroban Phase 2 & 3 Roadmap

Phase	Contracts	Key Primitive	Timeline
Phase 1	AutoInvestVault | PreferenceStore	Event-driven per-user invest_bps | off-chain bridge	T3 | This round
Phase 2	InvestRouter | DSE Adapter | VaultFactory	Full on-chain cross-contract | no user action	Next SCF round
Phase 3	YieldVault | Blend/DeFindex adapters	Settlement window yield | principal-protected	SCF round +2

Phase 3 converts InveStar corridor into real utility-driven Stellar DeFi TVL - backed by actual payment flows, not speculative mining.

Section 9 - System State Machine & 16-Week Timeline

▶️ End-to-End Transaction State Machine

```
INITIATED -> AUTHENTICATED -> USDC_ON_CHAIN (SCP 3-5s)
  +-(Track 1)-> SEP24_FLOW -> CASH_PICKUP_ISSUED
  +-(Track 2)-> TREASURY_FLOW -> USDC_CONVERTED -> BDT_INITIATED -> BDT_DELIVERED
     (both tracks converge)
INVEST_CHECK
  +-(toggle ON)-> DSE_TRADE (B2B DSE OMS live | Soroban AutoInvestVault T3: on-chain automation) ->
COMPLETE
  +-(toggle OFF)-----------------------------------------------------------------> COMPLETE
```

16-Week Delivery Timeline

Milestone	Deliverable	Weeks	Cost
TRANCHE 1 - MVP	Stellar wallet + SEP-10/24 on testnet	1-6	$43,600
M1.1	AES-256-GCM/HSM | Horizon API | pooled accounts | testnet	1-4	$23,800
M1.2	SEP-10 + SEP-24 MoneyGram Ramps (confirmed partner)	1-3	$17,850
T1 QA	E2E tests | QA report | zero P0/P1	6	$1,950
TRANCHE 2 - TESTNET	Treasury + MG Transfer + SEP-12 + 50-user beta	5-11	$45,225
M2.1	USDC->USD | MoneyGram FX | FX guardrails | idempotency	5-7	$16,800
M2.2	MG Transfer API | bKash + bank + cash | all 3 types | retry	5-7	$14,400
M2.3	SEP-12 KYC | AML checklist	7-8	$5,250
M2.4	50 beta users | monitoring | Stellar community demo	9-10	$6,500
T2 QA	Full SEP + MG regression | >=80% coverage	11	$2,275
TRANCHE 3 - MAINNET	Mainnet + remit-to-invest + Soroban + mobile + SDK	10-16	$52,500
M3.1	All services MAINNET | Azure HSM prod | Grafana | DR/HA	10-11	$11,900
M3.2	Stellar settle -> DSE OMS | >=3 test trades	10-12	$16,800
M3.3	Soroban AutoInvestVault | MIT licence | testnet tx hash	11-12	$5,950
M3.4	React Native | iOS TestFlight + Android Play Store	11-12	$10,500
M3.5	>=5 live mainnet txs | FX alerts | BD 2.5% reconciliation	13	$2,625
M3.6	NPM SDK | hosted docs | external dev tester	14	$2,450

T3 QA	>=500 VU | p95<3s | real device testing | playbook	15-16	$2,275

Section 10 - Technology Stack

Backend	Stellar	Infrastructure	Frontend
NestJS + TypeScript	Stellar TypeScript SDK	Azure Key Vault HSM FIPS 140-2 L3	Next.js | Web app | live
PostgreSQL (Supabase RLS)	Horizon API + SSE streaming	AES-256-GCM | Supabase edge fn	React Native | Mobile | T3 | new
Redis + Bull Queue	SEP-10 / SEP-24 / SEP-12	Azure APIM | rate limiting	React | DSE OMS UI | live
Supabase Edge Functions	USDC SAC | Circle-issued	GitHub Actions | CI/CD + Jest	Capacitor | mobile packaging
Docker | Rust | Soroban	Soroban SDK (Rust no_std)	Grafana / BetterUptime (T3)	TanStack Query | server state

==========

ANNEX A-D Supporting Evidence

Traction Evidence | Budget | Full Milestone Detail | Regulatory & Open-Source

Annex A - Traction Evidence - Verifiable In-Submission

> All traction verifiable without leaving the submission. Every item has a live URL or Horizon API endpoint.

A1. On-Chain Stellar Accounts - 8 Verified (snapshot 2026-04-18)

InveStar operates 8 Stellar accounts across testnet and mainnet. All verifiable on Stellar Expert. Testnet resets 2-4 times/year (SDF policy) - demo video captures live tx hashes as permanent proof.

Rank	Account Address (truncated)	Role	Network	Activity / Metrics	Status
1	GAIH3ULLFQ4D...K3QJZNSR	Primary custodial hub	TESTNET	1,000+ ops · 993 accounts created on-chain · continuous	#15607A
2	GD52OJS2I2EM...HT7B5	Beta testnet account	TESTNET	7+ payments · active user beta flows	#1A6B55
3	GCGH5TLXIMBE...GECPJ	Auth / SEP services	TESTNET	Active · SEP-10/24 operations	#1A6B55
4	GBYFBWPV3...IOZOSSI	stellar-ramp-service Mod A	TESTNET	Backend signing · Horizon API calls	#5A8A6A
5	GAGZEFSXEER5...7XVNM	SEP-10/24 ramp account	TESTNET	Ramp flow ops · anchor interaction	#5A8A6A
6	GCCEMCYSJLY7...O67IN	Treasury / USDC Mod C	TESTNET	Conversion ops · FX guardrails	#5A8A6A
7	GAIH3ULLFQ4D...K3QJZNSR	Primary wallet	MAINNET	3 ops · 1 payment (beta-limited)	#C8963E
8	GBTSIUX65PKO...WSOG	Secondary mainnet	MAINNET	Active (beta-limited)	#C8963E

★ *Ordered by testnet operation count · Mainnet accounts intentionally limited during closed beta · Mainnet cutover Q2 2026*

Figure 6: Stellar accounts ranked by testnet operation count | All verifiable on stellar.expert

TESTNET: 1,000+ ops | 993 beta-user wallets created on-chain:

#1 GAIH3ULLFQ4DGSECF2AR555KZ4KNDGEKN4AFI4SU2M7B43MGK3QJZNSR - 1,000+ ops | 993 wallets
#2 GD52OJS2I2EMOCRJCCCA6ES5V67RUO7DH3Y77G4ZGN447G6DAB5HT7B5 - 7+ payments | active
#3 GCGH5TLXIMBETOG6GP2ZGBR4ONYGUHWYSA66L7PG32XR7ZH762EGECPJ - Auth/SEP-10/24 ops
#4 GBYFBWPV3QB2BAH6HOJG4G7J5SQFNWU2CSV3CIEGSBQFH5RDKIOZOSSI - stellar-ramp-service
#5 GAGZEFSXEER5ZDIAXLEHJ7MFTZRTYWPCKU3SBQI6Z5GJLIYCBMY7XVNM - SEP-10/24 ramp
#6 GCCEMCYSJLY7YRPQ4WJMXDUP3RNCC7F2XOXZAYVULSWPGZWRB5PO67IN - Treasury/USDC

MAINNET: beta-limited | cutover Q2 2026:

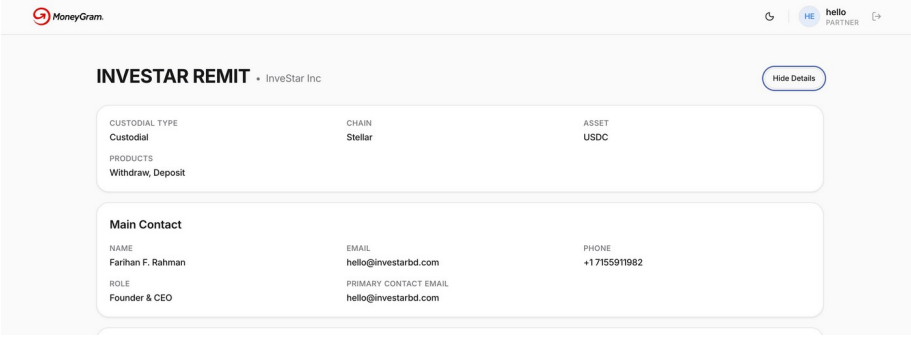
#7 GAIH3ULLFQ4DGSECF2AR555KZ4KNDGEKN4AFI4SU2M7B43MGK3QJZNSR - 3 ops | 1 payment
#8 GBTSIUX65PKO6VMUX4QQH37YKVY47OLLFWYCAS5NGBBJ4SCQ3CK2WSOG - Active

```
# Primary hub - 1,000+ ops, 993 wallets (testnet)
curl
"https://horizon-testnet.stellar.org/accounts/GAIH3ULLFQ4DGSECF2AR555KZ4KNDGEKN4AFI4SU2M7B43M
GK3QJZNSR/operations?limit=200&order=desc"
# Mainnet
curl
"https://horizon.stellar.org/accounts/GAIH3ULLFQ4DGSECF2AR555KZ4KNDGEKN4AFI4SU2M7B43MGK3QJZNS
R/operations?limit=200&order=desc"
```

Live traction dashboard: https://app.investarbd.com/traction
Demo wallet (no signup): https://app.investarbd.com/wallet-demo
Demo video with live tx hashes: https://www.youtube.com/shorts/NuoyWrr63Qg

A2. MoneyGram - Confirmed Partner Registration (Exhibit MG-1)

> NOT a letter of intent. InveStar is registered in MoneyGrams partner management system with Stellar USDC Withdraw and Deposit products ACTIVE. Directly confirms M1.2 (SEP-24 Ramps), M2.2 (Transfer API), Module C FX infrastructure.



MoneyGram Partner Portal - INVESTAR REMIT | Chain: Stellar | Asset: USDC | Products: Withdraw, Deposit | Status: Active registered partner

Portal Field	Verified Value
Company	INVESTAR REMIT | InveStar Inc
Chain	Stellar confirmed
Asset	USDC confirmed
Products	Withdraw, Deposit - ACTIVE
Main Contact	Farihan F. Rahman (Founder & CEO) | hello@investarbd.com
Status	Active - Registered partner in MoneyGram partner management system

Google Doc: https://docs.google.com/document/d/1mYU8OJoP9czHEVRUqrLV5YODS8GxGuWOaY3mvYeOU-g/edit | MG Ramps sandbox: https://extcloudtest.moneygram.com

A3. 9 Capabilities Shipped - Live Today

Capability	Description	Proof
Stellar SDK	993 wallets | 1,000+ ops | live Horizon data	app.investarbd.com/traction
Custodial Wallets	993 on-chain | AES-256-GCM key custody	supabase/functions/create-stellar-wallet
DEX Swaps	pathPaymentStrictSend | slippage protection	supabase/functions/stellar-dex-swap
Fiat Ramps	SEP-24 workflow shipped: auth, wallet linkage, fees, tx records, interactive URL handoff. Live anchor endpoints = M1.2 deliverable.	supabase/functions/stellar-anchor-ramp
Remittance	USDC->BDT/INR/PHP | BFIU compliance	src/pages/SendMoney.tsx
Mobile Apps	Capacitor native Android + iOS | biometric-ready	capacitor.config.ts | /android
DSE OMS	DSE-licensed | only startup with direct DSE integration	oms-investar.dev.sandbox3000.com
Traction Dashboard	Live Horizon data + reviewer curl	app.investarbd.com/traction

	commands	

A4. Team - Corridor Domain Experts

Name	Role	Credentials
Farihan Rahman	CEO since 2023	EIR Draper University | Nasdaq Milestone Makers Fall 2025 (first BD EdTech/FinTech) | O-1 + EB-2 NIW
Shafkat Alam	CTO since mid-2023	University of Toronto CS | Leads all Stellar architecture + production code
Dipro Chowdhury	COO since mid-2023	Masters MEng University of Bristol | 10+ years capital markets investor

Market: US-BD corridor $26-30B/yr | $32-33B record 2025 | 6.49% avg cost (WB Q1 2025) | InveStar target ~1% | BD govt 2.5% incentive on all compliant transfers

Phase 2: MENA->SEA - 40M+ workers | $100B+/yr | same Stellar+MG architecture | no rebuild needed

Annex B - Budget Justification

Tranche	Focus	Cost	%	Weeks
T1 MVP	Stellar wallet + SEP-10/24 on testnet	$43,600	30.9%	1-6
T2 Testnet	Treasury + MG Transfer + SEP-12 + 50-user beta	$45,225	32.0%	5-11
T3 Mainnet	Mainnet + remit-to-invest + Soroban + mobile + SDK	$52,500	37.1%	10-16
TOTAL	Back-weighted | $8,675 under $150K cap	$141,325	100%	<=16

Payment	Trigger	%	Amount
#0 Upfront	Award acceptance	10%	$14,132.50
#1 MVP	All T1 milestones verified	20%	$28,265.00
#2 Testnet	All T2 milestones verified	30%	$42,397.50
#3 Mainnet	All T3 milestones verified	40%	$56,530.00

Role	Rate	Justification
Senior dev x2	$80-85/hr	University of Toronto CS + Bristol MEng. Stellar TypeScript SDK, AES-256-GCM/HSM, NestJS. Toronto/UK market rate.
Mid dev x1	$65-75/hr	BD-based. React Native mobile, integration testing, QA tooling, SDK documentation.

Annex C - Deliverables: Full Milestone Detail

Tranche 1 - MVP - $43,600

Milestone	Scope	Verifiable Proof
M1.1 $23,800	stellar-ramp-service: NestJS + Stellar SDK + Horizon. AES-256-GCM/HSM key custody. 993 wallets.	1 Testnet address + tx hash 2 Key custody verified 3 GitHub JIT signing 4 Horizon Postman
M1.2 $17,850	SEP-10 + SEP-24 MoneyGram Ramps (confirmed). Full withdrawal flow + allowlist.	1 Demo video SEP-10->SEP-24->USDC->ref 2 Testnet tx hash 3 JWT log 4 Exhibit MG-1
T1 QA $1,950	E2E tests. QA report. Zero P0/P1.	1 QA matrix M1.1+M1.2 2 Zero P0/P1 3 Public URL + SEP-24 triggerable

Tranche 2 - Testnet - $45,225

Milestone	Scope	Verifiable Proof
M2.1 $16,800	USDC->USD via MoneyGram FX. FX guardrails | idempotency | Pre-Payment Disclosure.	1 100% FX unit tests 2 E2E USDC->USD log 3 Idempotency: dup rejected 4 Supabase demo
M2.2 $14,400	MoneyGram Transfer API. All 3 payout types. Retry logic.	1 Sandbox $100 test 2 Reference logs 3 Retry video 4 GitHub module
M2.3 $5,250	SEP-12 KYC with MG anchor. AML checklist + data-retention.	1 SEP-12 demo video 2 AML checklist 3 GitHub commit 4 Audit Bank confirmed
M2.4 $6,500	Full testnet. 50 beta users. Monitoring. Stellar community demo.	1 50+ sessions 2 Monitoring dashboard 3 Community post 4 Feedback report
T2 QA $2,275	Full regression SEP-10/24/12 + MG Transfer. >=80% coverage.	1 Test report 2 Coverage >=80% 3 Zero P0

Tranche 3 - Mainnet - $52,500

Milestone	Scope	Verifiable Proof
M3.1 $11,900	All services to MAINNET. Azure HSM prod. Grafana. DR/HA.	1 Mainnet tx hash 2 Grafana URL 3 DR failover report 4 Alert demo
M3.2 $16,800	Horizon SSE -> DSE OMS. Stellar settle -> invest. >=3 test trades.	1 Live Stellar->DSE video 2 >=3 trades 3 Mainnet tx hashes 4 GitHub
M3.3 $5,950	Soroban AutoInvestVault P1. MIT licence. Testnet deployment.	1 GitHub public + MIT 2 Testnet tx hash 3 README + invocation guide
M3.4 $10,500	React Native: wallet + SEP-24 + biometric. TestFlight + Play Store.	1 iOS TestFlight USDC send 2 Android Play Store 3 Demo video 4 GitHub
M3.5 $2,625	Stellar tx alerts | FX guardrail | BD 2.5% reconciliation. >=5 live mainnet txs.	1 >=5 mainnet tx hashes 2 FX alert screenshot 3 MG mainnet approval
M3.6 $2,450	TypeScript/JS SDK. NPM + hosted docs + quickstart.	1 NPM versioned 2 Docs live 3 Dev tester verified
T3 QA $2,275	Mainnet regression. >=500 VU p95<3s. Real devices. Incident playbook.	1 >=500 VU p95<3s 2 Device QA 3 Zero P0 4 Playbook

Annex D - Regulatory Constraints & Open-Source Plan

Constraint	Architecture Impact
Bangladesh Bank USDC prohibition	Module C mandatory - USDC converted to USD outside Bangladesh via MoneyGram FX before any BDT delivery. Direct USDC receipt blocked at code level.
MoneyGram Pre-Payment Disclosure	Enforced in Module B before every Transfer API call. Cannot be bypassed in retry flows.
BD govt 2.5% remittance incentive	M3.5 reconciliation tooling calculates and logs the bonus per transaction.
SCF Soroban open-source mandate	Soroban AutoInvestVault (T3 planned) published under MIT licence at M3.3. Hard T3 precondition. Current remit-to-invest uses live B2B DSE OMS. Phase 2+3 contracts also committed to future OSS.

Open-Source Commitment	Detail
AutoInvestVault + PreferenceStore MIT OSS	Phase 1 contract, README, arch doc, MIT licence committed before T3 payment claim. Testnet tx hash as proof.
Public TypeScript SDK NPM	SDK wrapping Stellar wallet, SEP-10/24/12, MG payout APIs. Versioned NPM. Hosted docs.
Phase 2+3 contracts MIT OSS (future rounds)	InvestRouter, VaultFactory, DSE Adapter, YieldVault - MIT licence in future SCF submissions.